



H2S

BHARATIYA ANTARIKSH HACKATHON 2026



Team Name : Web Hackers

Team Leader Name : Amaan Khan

Problem Statement : Challenge 4 - Route Resilience: Occlusion-Robust Road Extraction & Graph-Theoretic Criticality Analysis for Urban Mobility

Team Members

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The Idea: Route Resilience

How is it different?

Most teams stop at road segmentation, a crowded, hard-to-win space. We go further: we heal occluded masks into a CONNECTED routable graph (MST + Disjoint-Sets), then apply graph theory to find single points of failure and stress-test the city. Occlusion-robust extraction + topological healing + disaster simulation in one pipeline.

How does it solve the problem?

We recover roads hidden under canopy, shadow and cloud, pinpoint the 'Gatekeeper Nodes' whose failure fragments the city, and simulate floods or accidents to quantify connectivity loss, rerouting and travel-time impact, turning raw imagery into planning and disaster-response decisions.

USP

An interactive Resilience Index: planners disable a Gatekeeper Node and instantly see the city fragment, with rerouting and travel-time effects. Engineering-grade network intelligence aligned with ISRO's NNRMS mandate, not just a prettier map.

Key Features

Occlusion-Aware Extraction

Transformer-based segmentation infers road continuity under canopy, shadows, vehicles and clutter, with high occlusion-recall.

Topological Reconstruction

Heals fragmented masks into a routable weighted graph using MST + Disjoint-Sets (Union-Find).

Structural Intelligence

Identifies 'Gatekeeper Nodes': high-betweenness single points of failure, plus bridges & articulation points.

Simulated Stress Testing

Predicts impact of floods or accidents: connectivity loss, rerouting and travel-time increase.

Quantitative Criticality Map

Spatial heatmap of critical intersections across urban, suburban and rural terrain.

Interactive Resilience Index

Disable a node and watch the city fragment live, with real-time resilience metrics.

How It Works: Process Flow



Inputs: Sentinel-2 / LISS-IV / Cartosat-3 imagery + OpenStreetMap, SpaceNet, DeepGlobe & OpenSatMap ground truth. Output: a Resilience Index on an interactive map. The graph analysis engine (orange) is already built and tested.

Interactive App Wireframe

MAP VIEW

Extracted road network over the city
Critical junctions highlighted in red
Toggle: imagery · roads · graph

▶ Simulate Collapse

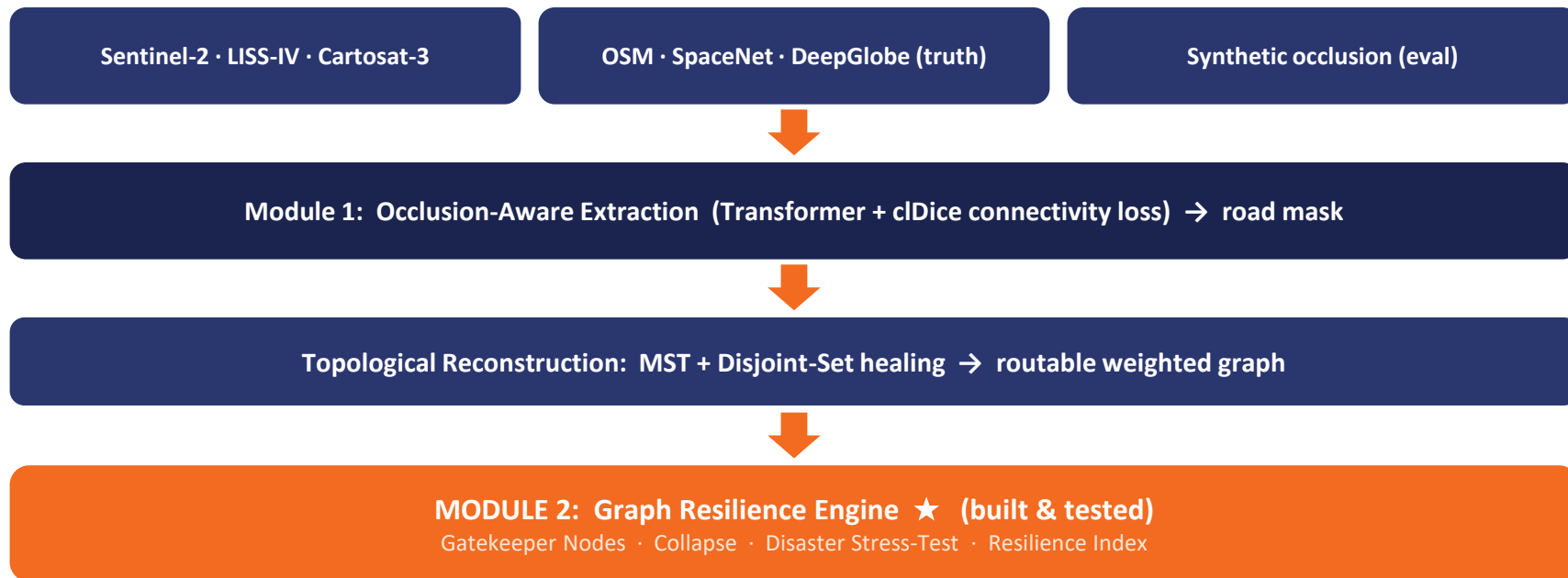
Scenario: Flood

Remove top-k critical nodes

Resilience metrics

Largest component: 56%
Network efficiency: -41%
Population isolated: 38,000
Hospitals cut off: 2

System Architecture



Aligned with ISRO's NNRMS mandate (MeitY · Ministry of Consumer Affairs); served as an interactive Resilience Index app.

Technology Stack

AI / Computer Vision

PyTorch · U-Net / UNet++ / DeepLabV3+ / Transformer · connectivity-aware cIDice loss · Albumentations, OpenCV

Graph & Analysis

NetworkX · PyTorch Geometric (GNN) · MST + Disjoint-Sets healing · betweenness (Gatekeeper Nodes) · pytest (33 tests)

Geospatial Data

Sentinel-2 · Resourcesat LISS-IV · Cartosat-3 · SpaceNet / DeepGlobe / OpenSatMap · OSM (OSMnx) · Rasterio / GDAL · FilFinder

App & Tooling

Interactive map (Leaflet.js / Mapbox) · Streamlit · QGIS, Matplotlib · Python · Git / GitHub

Estimated Implementation Cost

Built almost entirely on free, open-source software and open public datasets, with minimal cost to build and pilot.

Cost item	Estimated cost
Open-source software & libraries	₹0
Public satellite & OpenStreetMap datasets	₹0
GPU compute for model training (cloud, one-time)	₹5,000 to ₹10,000
Web-app hosting (demo)	₹0 to ₹1,000 / mo

Pilot-ready at a fraction of commercial GIS costs.

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THANK YOU